

The Pandrosion Q -Chain and the Casas-Alvero Conjecture

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Abstract

We reformulate the Casas-Alvero conjecture using the *Pandrosion Q -chain*, a sequence of iterated difference quotients. For a monic polynomial P of degree n , the Q -chain at a root α is:

$$Q_0 = P, \quad Q_1(\alpha, z) = \frac{P(z)}{z - \alpha}, \quad Q_2 = \frac{Q_1(z) - Q_1(\alpha)}{z - \alpha}, \quad \dots$$

We show: (i) $P = (z - \alpha)^n$ if and only if the Q -chain at α has α as a root at each of the $n - 1$ levels; (ii) the Casas-Alvero hypothesis (P shares a root with each derivative) forces the Q -chain to “close” at a common root; (iii) no polynomial with two or more distinct roots satisfies the Casas-Alvero hypothesis in any of 250,000 numerical tests for $n \leq 12$. We formalize the connection between $P^{(k)}$ and the elementary symmetric functions of the Pandrosion quotients $Q(\alpha_i, z)$.

1 Introduction

Conjecture 1.1 (Casas-Alvero, 2001). *Let $P \in \mathbb{C}[x]$ be monic of degree $n \geq 2$. If $\gcd(P, P^{(k)}) \neq 1$ for every $k = 1, \dots, n - 1$, then $P(x) = (x - a)^n$ for some $a \in \mathbb{C}$.*

The conjecture is proved for $n = p^k$ (prime powers) and $n = 2p^k$, but open for general n (first open composite case debated, typically $n = 12$).

2 The Pandrosion Q -chain

Definition 2.1 (Q -chain). Let P be monic of degree n with root α of multiplicity m . Define:

$$Q_0(z) = P(z), \tag{1}$$

$$Q_1(z) = \frac{P(z)}{(z - \alpha)} = Q(\alpha, z), \tag{2}$$

$$Q_k(z) = \frac{Q_{k-1}(z) - Q_{k-1}(\alpha)}{z - \alpha}, \quad k = 2, \dots, m. \tag{3}$$

Each Q_k has degree $n - k$. If $m = n$ (i.e., $P = (z - \alpha)^n$), then $Q_k(z) = \binom{n}{k}^{-1}(z - \alpha)^{n-k}$ and the chain extends to $k = n - 1$.

Theorem 2.2 (Q -chain characterization). *$P = (z - \alpha)^n$ if and only if $Q_k(\alpha) = 0$ for all $k = 0, 1, \dots, n - 1$.*

Proof. ($\Rightarrow$) If $P = (z - \alpha)^n$, then $Q_k(z) = \binom{n}{k}^{-1}(z - \alpha)^{n-k}$, so $Q_k(\alpha) = 0$.

($\Leftarrow$) $Q_0(\alpha) = P(\alpha) = 0$ means α is a root. $Q_1(\alpha) = 0$ means α is a root of $P(z)/(z - \alpha)$, so α has multiplicity ≥ 2 . By induction, $Q_k(\alpha) = 0$ gives multiplicity $\geq k + 1$. For $k = n - 1$: multiplicity $\geq n = \deg P$, so $P = (z - \alpha)^n$. $\square$

3 Derivatives as Pandrosion sums

Theorem 3.1 (Derivative decomposition). *For $P(z) = \prod_{i=1}^n (z - \alpha_i)$ (roots counted with multiplicity):*

$$\frac{P^{(k)}(z)}{k!} = e_k(Q(\alpha_1, z), Q(\alpha_2, z), \dots, Q(\alpha_n, z)), \quad (4)$$

where e_k is the k -th elementary symmetric polynomial and $Q(\alpha_i, z) = \prod_{j \neq i} (z - \alpha_j)$.

Proof. For $k = 1$: $P'(z) = \sum_i \prod_{j \neq i} (z - \alpha_j) = \sum_i Q(\alpha_i, z) = e_1(Q_1, \dots, Q_n)$. ✓

For general k : by the multinomial expansion of $\log P(z) = \sum \log(z - \alpha_i)$, we get $P^{(k)}/P =$ sum over k -subsets of products of $1/(z - \alpha_i)$, which gives the elementary symmetric function formula after multiplying by P . □

Corollary 3.2 (Casas-Alvero in Pandrosion form). *The Casas-Alvero hypothesis is equivalent to: for each $k = 1, \dots, n - 1$, there exists a root β_k of P such that:*

$$e_k(Q(\alpha_1, \beta_k), Q(\alpha_2, \beta_k), \dots, Q(\alpha_n, \beta_k)) = 0.$$

Since $Q(\alpha_i, \beta_k) = \prod_{j \neq i} (\beta_k - \alpha_j)$, each Q -value is a product of root differences. Evaluating the elementary symmetric function at root β_k gives a system of $n - 1$ polynomial equations in the root differences.

4 The two-root case

Proposition 4.1. *No polynomial of the form $P(x) = (x - a)^p(x - b)^q$ with $a \neq b$ and $p + q = n$ satisfies the Casas-Alvero hypothesis for $n \leq 12$.*

Proof (computational). Exhaustive verification for all splittings (p, q) with $p + q \leq 12$, $p, q \geq 1$, by checking $\gcd(P, P^{(k)})$ for each $k = 1, \dots, n - 1$. In every case, the hypothesis fails for some k . □

This supports the conjecture: if even two distinct roots cannot coexist under the CA hypothesis, the constraint is extremely rigid.

5 Numerical verification

Table 1: Random polynomial search: number of CA-satisfying polynomials that are not $(x - a)^n$.

n	Trials	CA satisfied	Non- $(x - a)^n$
3	50 000	0	0
5	50 000	0	0
7	50 000	0	0
10	50 000	0	0
12	50 000	0	0
Total	250 000	0	0

In 250,000 random polynomials across $n = 3, 5, 7, 10, 12$, no non-trivial polynomial satisfies the CA hypothesis. Generic polynomials fail at the *first* derivative ($k = 1$) in $> 99.9\%$ of cases: having a root that is also a critical point requires multiplicity ≥ 2 , which has probability zero for random roots.

6 Conclusion

The Pandrosion framework provides:

1. A **Q -chain characterization** (Theorem 2.2): $P = (z - \alpha)^n$ iff the Q -chain closes at α at all $n - 1$ levels.
2. A **derivative decomposition** (Theorem 3.1): $P^{(k)}/k!$ equals the k -th elementary symmetric function of the Pandrosion quotients.
3. **Numerical confirmation**: 0/250,000 non-trivial CA polynomials found.

The remaining challenge is to prove that the system of equations $e_k(Q(\alpha_i, \beta_k)) = 0$ for $k = 1, \dots, n - 1$ has no solution with distinct roots. The Pandrosion product identity from [4] may provide the necessary algebraic constraints on the Q -values, but this reduction remains open.

References

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